A Dual-Model Phase Vocoder for Transparent Time-Warping under Manual Control

warptempo

Engineering note: the perceptual findings are one expert listener's judgments, reported as such. Claims are stated at the scope where they hold.

Abstract

We describe a phase vocoder for high-ratio time-warping of recorded music under manual control, in which audible artifacts are minimized and contained rather than absent in principle. The engine is built on a faithful implementation of Prusa-Holighaus phase-gradient heap integration (PGHI, "heap"), with classical Laroche-Dolson identity phase-locking ("peak") implemented as a selectable per-segment alternative on the same short-time Fourier transform (STFT) substrate. The central engineering result is that **neither model is sufficient on its own** for transparent time-warping of this material — each has a distinct failure mode the other handles well — but that the two, made selectable per segment and seeded identically at shared phase-reset points, together produce output more transparent than any general-purpose time-stretching tool we evaluated. If forced to a single model, peak is the one we would keep: it is dull but stable everywhere, whereas heap alone is distractingly unstable on the exposed passages that orchestral writing is full of. Peak is therefore the dependable default; heap is applied where the surrounding texture provides enough masking for its residual instability to disappear, which is most dense and loud writing, and where it repays that masking with a spectral life peak cannot recover. The complementary failure modes are a property of the two models rather than of any particular material; the demonstrations here use the 2024 Christopher Bernauer / Decca Eloquence remasters of the 1972 Krips Mozart symphony cycle, warped toward historically-informed tempos. Orchestral material is chosen because it is the most demanding masking environment for this class of algorithm — exposed lines, natural-acoustic recording, large hall — and a tool that handles it transparently has headroom for genres with denser masking. The source is best-in-class recording quality, which sharpens rather than weakens the transparency test: there is detail worth preserving, and the criterion is whether it survives the warp. The perceptual findings are one expert listener's judgments accumulated over many iterations, not a formal listening study.

---

1. Motivation

The task is high-ratio time-warping of recorded music in which the audible artifacts introduced by the algorithm are minimal and perceptually contained — the output should differ from the source only in tempo, to the extent that the constraints of the method permit. Phase resets are themselves artifacts by construction (the synthesis phase is discontinuously re-seated to the analysis phase at each marker); the discipline is to place them where the surrounding material masks them. The work was developed on the 2024 Christopher Bernauer remasters (Decca

Eloquence) of the 1972 Krips Mozart symphony cycle, warped toward historically-informed tempos. That material is used for the demonstrations below; the source is widely regarded as exemplary recording quality, and the 2024 remasters are best-in-class restorations of it. We name this because the transparency criterion is more demanding on high-quality source: there is more detail to preserve, and the warp has less room to hide.

The choice of orchestral material is not arbitrary: it is the most demanding masking environment for phase-vocoder time-warping. The size of a symphony orchestra and the natural-acoustic recording technique typical of the genre — sparse microphone counts, large hall acoustics, minimal close-miking — preserve substantial silence and air around individual instrumental lines and capture extended exposed solo passages, both of which give algorithmic artifacts maximal room to be heard. Genres with denser sustained spectra and more pervasive percussive activity (rock, pop, electronic dance music) provide stronger masking; the author's two decades of producer practice on such material support the expectation that the technique generalizes favorably to those genres. We make this expectation explicit here and present it as such — practitioner judgment, not formal evaluation — because the asymmetry runs in the direction that strengthens the claim: a tool that handles the most artifact-revealing case transparently has headroom for material that masks more, not less. The technique is therefore not specific to orchestral material — the complementary-failure-mode result in Section 4 is a property of the two phase-correction models, not of the repertoire. What *is* specific is the workflow: the operator authors phase-reset markers by hand, which is manual work (less of it than one might expect — see Section 2 — but manual). The method is therefore offline and operator-driven rather than real-time and automatic, and that is the only constraint on where it applies.

This separates it from general-purpose time-stretchers — the open-source and commercial tools surveyed in Section 4.6 — which are engineered for real-time operation, arbitrary input, formant-preserving vocals, low latency, and zero-configuration robustness on material the developer has never heard. Those properties cost transparency. A tool that spends that budget on transparency instead, accepting offline operation and manual authoring, is the natural design when the goal is fidelity rather than throughput.

The classical phase vocoder's difficulty is well known and is laid out in detail by Laroche and Dolson [1]: time-scaling by using different analysis and synthesis hops requires correcting the STFT phase, and naive correction destroys *vertical* phase coherence — the phase relationship across frequency bins within a single frame — producing the family of artifacts collectively called phasiness (reverberation, loss of presence, transient smearing). Laroche-Dolson's identity phase-locking restores vertical coherence by picking spectral peaks and locking each peak's region of influence to the peak's phase. Prusa and Holighaus [2] take a different route: they estimate *both* partial derivatives of the STFT phase — in time and in frequency — and integrate them adaptively, enforcing horizontal and vertical coherence without any peak-picking or transient detection.

Both methods work. The problem this paper addresses is that on our material, each one works *in a different place*, and the difference is large enough to hear.

2. Architecture

The engine is an in-process C++ phase vocoder built on a faithful, centered Prusa-Holighaus PGHI substrate, with Laroche-Dolson identity phase-locking implemented as a selectable per-segment alternative on the same STFT grid. The signal path is: analysis STFT $\rightarrow$ per-frame phase correction (heap or peak, per segment) $\rightarrow$ overlap-add resynthesis $\rightarrow$ inline limiter.

Two orderings, opposite directions. It is worth stating plainly that the implementation and the authoring philosophy run in opposite directions, and that this is deliberate. *In implementation,* PGHI is the foundation and Laroche-Dolson is built onto it: the STFT substrate is the one PGHI requires — origin-centered window, $M = 2N$ zero-padded transform — and peak runs on that same grid (see "Peak on the shared grid" below). This is the correct order for the code, because the gradient-integration method is sensitive to the STFT convention in a way peak is not. *In authoring,* the relationship is reversed: peak is the stable, safe, dull default, the model an operator can apply anywhere without distraction, and heap is applied liberally for spectral life and excitement where the texture supports it. We build the code PGHI-first because that is the correct substrate; we author peak-first because it is the dependable default.

Faithful STFT setting. The analysis uses a 4096-sample Hann window and an $M = 8192$ ($= 2N$) zero-padded transform, origin-centered. This is, to within four samples of window length, the configuration of the published PGHI evaluation (a 4092-sample Hann window, $M = 8192$). The four-sample window difference is the only deviation from the published STFT setting, and it is not a meaningful one: $N = 4096 = 2^{12}$ is FFTW-clean and the difference is far below any perceptual or numerical consequence. We note it for completeness rather than as a divergence of substance.

Fixed synthesis hop, variable analysis hop. The synthesis hop is fixed at $R_s = N/4$. The analysis hop is computed per-frame from the local stretch ratio, $R_a = R_s/\alpha$, so the time grid the operator authors is realized by *varying where we read* rather than where we write. This is the right way around for a warp: the output is laid down on a uniform grid, and tempo variation lives entirely in the analysis-side sampling. The phase-derivative estimates accommodate the variable hop directly (Section 3); a tempo change across a warp marker needs no special-casing, because each finite difference is normalized by the actual local hop on its own side.

Peak on the shared grid. Laroche-Dolson peak-picking and phase-locking run on the same origin-centered $M = 2N$ grid PGHI uses, rather than on the native-length, un-padded transform of textbook LD. This is a deviation from canonical LD, and it is intentional: it lets both models share one analysis substrate with no second transform, no length mismatch, and no per-model latency seam. It is also, to anyone familiar with the method, an unsurprising one. Peak-picking is scale-relative — a local-maximum test is indifferent to how finely the frequency axis is sampled, and zero-padding interpolates the existing transform rather than fabricating maxima, so the same partials are found, located more precisely. The phase-lock transcribes the analysis inter-bin phase relationships verbatim rather than reconstructing them, so it carries no integration that the denser grid or the centered convention could perturb. Empirically, peak on the shared grid reconstructs the input to within numerical noise at $\alpha = 1$ (a null on the order of -224 dB), confirming that the grid change does not perturb it. We regard this as a property of LD's robustness, not as a result requiring defense.

FFT size. `N = 4096`. The dual-model split is what makes this size viable. Under Laroche-Dolson alone, a window this large smears transients and vertical structure, and the engine had to run at a smaller `N = 2560` to hold attacks and high-frequency detail together. Once the heap model carries the dense material — where the larger window's finer frequency resolution is an asset rather than a liability — and the peak model is retained per-segment for the exposed and transient material, the larger global `N` becomes the practical default. The size choice and the model split are not independent decisions; the split is the precondition for the size. (Window length remains an operator-set time-versus-frequency tradeoff: a shorter window favors transient clarity and upper-band articulation, a longer one favors low-frequency body and the gradient integration's frequency resolution. `N = 4096` is the faithful default; the choice is a judgment, not a forced setting.)

Two orthogonal marker collections. The operator authors two independent layers.

Warp markers shape the timemap — they set and inherit tempo, lock tempo across recapitulated material via labels, and define the entire stretch trajectory. This layer is where the musical expertise lives when the work calls for non-constant tempo, and it is deliberately not the subject of this paper. The classical phase-vocoder use case — one constant stretch ratio for an entire file — requires no warp-marker interaction at all: a single mandatory warp marker is present at the file's first frame with tempo 1.00, and the global stretch ratio is set entirely through a settings field (`scale`) with six-decimal precision. The marker need not be touched. Warp markers become an authoring surface only when the operator wants phrase-level tempo variation; in the classical-PV sense, the timemap demands zero expertise. The expertise the tool demands, when it demands any, is concentrated in this warp layer on the question of *where the tempo should go*.

Phase-reset markers are the subject of this paper. Each is anchored to a transient and carries three fields: a time, a disabled flag, and a **mode**. At a phase-reset marker the synthesis phase is re-seated to the analysis phase ($\theta = \varphi$), discarding accumulated phase history — a phase-only operation that never alters the stretch, which remains the timemap's job everywhere. The mode field selects, for the *segment* of audio governed by that marker, which phase-propagation model runs:

- **peak** — Laroche-Dolson identity phase-locking. The stable default.
- **heap** — Prusa-Holighaus phase-gradient heap integration (PGHI).
- **pass** — inherit the mode of the previous mode-owning marker (mirroring the way warp markers inherit tempo), resolving by a backward walk to a concrete peak/heap before the engine runs; peak when nothing precedes.

This layer demands far less expertise than the timemap. Phase-reset placement is locally scoped and transient-anchored — the markers go roughly where the onsets are — and under the heap model it tolerates *sparser* placement than peak did, because PGHI enforces vertical coherence without needing a reset to re-establish it as often. The tool pushes the operator's effort toward the musical decision and absorbs the signal-processing decision into the engine.

3. The two derivative axes

3.1 Notation

Following Prusa-Holighaus [2], write the discrete STFT phase as $\varphi(m, n)$ for frequency bin m and frame n, and the principal-argument (phase-unwrapping) operator as

$$[x]_{2\pi} = x - 2\pi \cdot \mathrm{round}\left(\frac{x}{2\pi}\right)$$

i.e. the wrap of x into $(-\pi, \pi]$. Let R_a be the analysis hop, R_s the synthesis hop, M the transform length, and b_a the analysis frequency step. Both papers build the corrected synthesis phase by *differentiating* the analysis phase along an axis, unwrapping the difference, and *integrating* the result on the synthesis side. The two axes — time and frequency — are treated separately, and the centered STFT convention determines what each axis must do.

3.2 The time axis

The classical phase-vocoder time derivative — the instantaneous-frequency estimate — is, per Laroche-Dolson [1] and in the centered form preferred by Prusa-Holighaus, computed by removing the *expected* linear phase advance of bin m before unwrapping, then adding it back:

$$(\Delta_t\varphi)(m,n) = \omega_m + \frac{\big[\,\varphi(m,n) - \varphi(m,n-1) - \omega_m R_a\,\big]_{2\pi}}{R_a}, \qquad \omega_m = \frac{2\pi m}{M}$$

The term $\omega_m R_a$ is the phase a stationary sinusoid at bin m's center frequency would accrue over one analysis hop. Subtracting it before $[\,\cdot\,]_{2\pi}$ is what makes the unwrap meaningful — it removes the bulk rotation so the principal argument captures only the small deviation from the bin's nominal frequency — and adding ω_m back afterward restores the true instantaneous frequency. The centered estimate normalizes each half-difference by its own actual hop (R_a backward, R_a forward), which is exactly what lets a stretch change across a warp marker pass through with no special handling. The time axis is identical in the classical PV and in PGHI; it is the half of the problem the classical PV already handles.

3.3 The frequency axis

The classical PV discards the frequency-direction derivative entirely, which is why it fails on anything but stationary sinusoids: a chirp or an impulse has real, non-negligible vertical phase structure that the time axis alone cannot see. PGHI restores it. In the origin-centered STFT convention — the convention the engine uses, and the one in which PGHI is specified — the analysis window carries no linear-in-bin group-delay term, so the expected per-bin progression on the frequency axis is *zero*. The frequency derivative is then simply the principal argument of the raw inter-bin difference:

$$(\Delta_f\varphi)(m,n) = \frac{\big[\,\varphi(m+1,n) - \varphi(m,n)\,\big]_{2\pi}}{b_a}$$

There is nothing to demodulate and nothing to re-add: the centered window is what makes the frequency axis this clean. This is the single most important convention decision in the implementation. A window applied *un-shifted* over $[0, N)$ — centered near sample $(N-1)/2$ rather than at the origin — would instead carry a linear-in-bin group-delay term on the analysis phase that the gradient integration would accumulate, and the published clean result would not be reproduced. The engine uses the centered convention precisely so that the frequency axis requires no special handling and PGHI behaves as published. We treat PGHI as a method to implement faithfully; the centered convention is part of that faithfulness, not a modification of it.

3.4 Integration: the heap

Given both derivatives, PGHI integrates them into a synthesis phase by a magnitude-prioritized flood fill. A max-heap holds bins with already-assigned phase as candidates to propagate *from*. The loudest available bin is popped; if it is a previous-frame bin, its phase steps forward in *time* to the current frame (trapezoidal rule on Δ_t); if it is a current-frame bin, its phase steps in *frequency* to its still-unassigned significant neighbors (trapezoidal rule on Δ_f). Each bin's phase is computed exactly once, and the order — loudest first — means coherence radiates outward from the spectral features that matter most. Significant bins are those whose magnitude exceeds a relative tolerance, $\texttt{abstol} = \texttt{tol} \cdot \max(s(\cdot, n) \cup s(\cdot, n{-}1))$ with $\texttt{tol} = 10^{-6}$; sub-tolerance ("quiet") bins are assigned a uniform-random synthesis phase, which is the published choice. We follow it exactly rather than substitute a variant. The bins this policy touches are those whose magnitude lies at or below the source recording's own quantization floor — bins carrying no meaningful content — so the phase assigned to them affects the reconstruction by an amount comparable to the floating-point reconstruction error of the transform itself (we use FFTW). The quiet-bin policy is, in that sense, beneath the numerical noise of the operation it belongs to.

The synthesis frequency step uses $b_s = \alpha\, b_a$ per the published method — the synthesis-side frequency resolution scales with the stretch factor, and the frequency integration scales with it accordingly.

At a phase-reset marker, and at the file's first frame, both models do the identical thing: seed $\theta = \varphi$ and skip integration. The models differ only in how they propagate *between* resets.

4. Findings

4.1 Each model's failure mode

The headline result, stated as one expert listener's judgment over many iterations on the demonstration material:

Peak (Laroche-Dolson) is the stable default. Without phase resets, peak alone produces output that is acceptable but dull. With phase resets, it substantially recovers vertical phase coherence at each reset point — but the inter-reset propagation still smears mid- and high-frequency content, attenuating perceptual brightness and apparent transient sharpness. Peak's failure mode is therefore not a transient artifact but a *steady-state perceptual loss*: stable dullness, reduced impact, reduced clarity in the upper bands. The output remains coherent and listenable everywhere; it lacks the perceptual presence of the source. Crucially, that loss is *uniform and undistracting* — it is the kind of degradation an ear adjusts to, not one it is repeatedly pulled toward. This is why peak is the model we would keep if forced to one.

Heap (PGHI) is the opposite failure mode. Heap preserves the spectral life that peak smears — the upper-band detail, the sense of impact at climaxes — but it is not artifact-free. It carries a low-level vibratory instability that is present throughout, and that becomes audible whenever the signal lacks the density to mask it. Where masking is sufficient — dense, loud, harmonically active writing — the instability disappears under the surrounding energy and heap is clearly preferable. Where masking is insufficient — exposed, quiet, or thinly-scored material — the instability is unmasked and distracting: a phase wobble on sustained tone that, to a listener attending to it, is enough to pull the ear away from the music. Heap as the sole mode is therefore unacceptable for classical material, where exposed writing is pervasive: if forced to a single model for this repertoire we would choose peak, accepting its dullness over heap's distraction. For genres with denser and more continuous masking (rock, pop, electronic dance music) heap alone is more plausibly usable, depending on instrumentation; even there, peak remains the dependable default and heap the model applied for effect. The reproduction of PGHI's published clean result on dense material, and the one place our experience diverges from it, are discussed in Section 4.6.

4.2 The criterion is masking, not dynamic

The single most practically important consequence of the above is that the choice between models is governed by *masking*, not by dynamic level and not by articulation per se. Heap is the right model wherever the surrounding texture is dense enough to mask its residual instability; peak is the right model wherever the texture is too exposed for that, regardless of how loud it is.

No exposed passage benefits from heap. Exposed material — quiet or not — has neither the spectral density nor the continuous energy to mask heap's instability, and peak's dullness is least audible precisely where the music is most exposed and least dense. Exposed and thinly-scored material is peak's.

Dense material is heap's. Loud, harmonically rich, continuously active writing provides the masking heap needs, and there heap's preserved presence is a clear gain over peak's smearing. This now covers a wider range of the music than an earlier, more conservative peak-default map assigned to heap; as the faithful convention reduced heap's residual instability, the share of material on which heap is the better choice grew accordingly.

Without heap, peak is insufficient at climaxes. The third point is the reason the dual-model architecture is required at all: peak alone produces a dynamically and spectrally flattened reading of the climactic, dense, harmonically-rich tutti passages that carry the dramatic weight of the music. A peak-only render is internally consistent — the dullness is uniform across the work — but consequently the passages that should be perceptually most prominent are rendered with the same reduced presence as everything else. Heap applied per-segment to those passages restores their perceptual prominence.

4.3 The authoring map

Because the choice is per-segment, the operator tags each section with the model that suits its texture, on the masking criterion of Section 4.2. The working map for the demonstration movement places heap on the dense and climactic writing — the prolonged tutti passages and the exposition close — and peak on the exposed, quiet, and thinly-scored material throughout, with the same assignments replicating across the exposition repeat and the recapitulation where the material recurs. As the faithful convention widened heap's usable domain, heap's share of the movement grew relative to the original peak-default map; the governing principle, however, is unchanged and is the one stated above: heap where density masks its residual, peak where the texture is too exposed. The map is visible directly in the authored marker files, where phase-reset markers carry `heap` tags on the dense and climactic entries and `peak` on the exposed material.

4.4 The model-handoff seam

The single remaining concern in the design is the seam where an outgoing heap segment meets an incoming peak segment, or the reverse. Both models re-seat $\theta = \varphi$ at the reset, so the handoff is not a discontinuity in the strict sense — but the overlap-add window blends two models' phase trajectories across the boundary, and the seam is in principle audible. In practice it is not audibly problematic on the demonstration material, for a structural reason: the authoring map (Section 4.3) places every mode switch on a *section boundary* — between exposed and dense writing, at the edges of the exposition close — which are points of strong dynamic and textural change and natural phase-reset locations independent of the mode-switch concern. Masking at section boundaries is correspondingly strong. A seam-crossfade refinement was designed in preliminary form and is held in reserve; the authoring placement to date has rendered it unnecessary. We classify it as not currently required rather than as outstanding work.

4.5 Operating envelope

The validated stretch envelope is modest by design. The nominal operating point is approximately a 1.5× speed-up; the largest *sustained* stretch on the demonstration movement is on the order of 1.3×. These ratios are gentle, and are not a sampling artifact concealing a wider claim: the objective is transparency, and transparency is a function of both material and ratio. Larger ratios are mechanically possible — the control surface admits arbitrary ratios — but the envelope reported here is the one we have evaluated and judged transparent on the demonstration material. Performance at larger ratios is unevaluated and no claim is made about it.

4.6 Comparison to evaluated tools, and relation to PGHI

We have, over the course of this work, compared the engine against a range of general-purpose time-stretchers and judged it preferable for transparent time-warping. Among open-source tools: Rubberband, Bungee, Signalsmith Stretch, and SoundTouch. Among commercial tools: zplane élastique (as deployed in Ableton Live and Reaper), Serato Pitch 'n Time (via Serato Sample), and Digital Performer's ZTX.

The claim, stated at its scope:

> *For transparent, high-ratio time-warping under manual control, this dual-model approach produced results we judged more transparent than every general-purpose tool we evaluated.*

The tools above are general-purpose: optimized for real-time use, formant-preserving vocals, low latency, and zero-configuration robustness on arbitrary input. The engine here is offline, hand-authored per-segment, and manual. The claim is that under those constraints transparency exceeds what we obtained from any general-purpose tool we evaluated; we do not claim it is a better general-purpose tool, and the constraints are stated as plainly as the comparison.

Relation to Průša & Holighaus [2]. With a faithful, centered implementation of PGHI we *reproduce* the published result on dense material: heap is clean and competitive there, exactly as the PGHI authors report. One divergence remains, and it is narrow: on *exposed, sustained* material heap retains an audible vibratory instability that makes it unacceptable as the sole mode for this repertoire. We attribute this to the test material rather than to a dispute about the method. The PGHI listening study used a relative comparison-category-rating criterion with the proposed algorithm as reference, and its excerpts — a drum solo, mixed pop, a solo-violin-with-castanets excerpt, a female vocal — do not include exposed, sustained orchestral strings. The nearest sustained-tone case in their set, a solo female voice, the authors themselves note is slightly more "phasy" than a competing tool; the solo-violin excerpt is overlaid with castanets and functions as a transient test rather than an exposed-sustained one. Our finding is therefore the expected direction: on exposed sustained material — a case the PGHI evaluation did not include — we hear an instability their test set would not have surfaced, and the dual-model exists to cover precisely that gap. We regard PGHI as a method this paper *implements and depends on*, not one it supersedes — heap is half of the result here, and the dual-model architecture only exists because PGHI gave us a model whose strengths are complementary to Laroche-Dolson's. We would welcome both the Laroche-Dolson and the Průša-Holighaus authors' independent evaluation of this work and of the demonstration material.

5. Limitations

- **Evaluation breadth.** The formal demonstrations are on one body of material (the 2024 Bernauer remasters of the 1972 Krips Mozart cycle) by one expert listener over many iterations, cross-checked informally with a small number of additional listeners. Orchestral material is the most artifact-revealing case for this class of algorithm; the author's two decades of producer practice on rock, pop, and electronic dance music support the expectation that the technique generalizes favorably to those genres, whose denser masking gives algorithmic artifacts less audible room. No formal cross-genre evaluation is presented here, and this is not a formal listening study.
- **Offline and manual.** The method requires hand-authored phase-reset markers and is not real-time. The authoring burden is modest — fewer markers than one might assume, and far less than the timemap demands — but it is manual, and that precludes drop-in or real-time use.
- **Heap is masking-dependent, not artifact-free.** Heap carries a low-level vibratory instability at all times; it is acceptable only where the surrounding texture masks it. Exposed, quiet, and thinly-scored passages do not, and are handled by tagging those segments peak. This is a genuine limit of the gradient-integration model on this material, not a solved problem — the dual-model contains it rather than eliminating it.
- **Window size is a tradeoff, not a solved optimum.** `N = 4096` is the faithful default and gives the gradient integration its finest frequency resolution and the fullest low-frequency body; a shorter window (e.g. `N = 2560`) trades some of that body for clearer upper-mid and top-end articulation. On this source — a 1972 recording whose low end is already a restoration compromise — the choice is a genuine judgment between two senses of faithfulness rather than a settled optimum, and the operator sets it.
- **Automated marker placement: declined, not infeasible.** Automated phase-reset placement is conceivable in principle and is a tractable engineering task — it would require classification of masked versus exposed material, both of which are within reach of established audio analysis methods. We have not pursued it. The judgment is that manual marker authoring produces measurably higher transparency than any automated

placement we would expect to achieve, and that the per-passage classification a competent operator performs by ear includes contextual information (phrase boundary, score knowledge, intended musical emphasis) not available to a signal-only classifier. We note the path here for completeness rather than as ongoing work.

6. Conclusion

The central result of this work is that **neither Laroche-Dolson identity phase-locking nor Prusa-Holighaus phase-gradient heap integration is sufficient on its own for transparent time-warping of this material; their per-segment combination is.** The two models exhibit complementary failure modes: peak (Laroche-Dolson) produces a steady-state perceptual loss — stable dullness and reduced upper-band clarity — uniform across all material, with phase resets recovering vertical phase coherence only at the reset points themselves; heap (PGHI) preserves the spectral life that peak smears but carries a low-level vibratory instability that is acceptable only where the surrounding texture masks it. If constrained to a single model, peak is the one to keep — its degradation is uniform and undistracting, whereas heap's is distracting on the exposed writing pervasive in this repertoire — and peak is correspondingly the engine's authoring default. The implementation is nonetheless built PGHI-first: the engine is a faithful, centered PGHI phase vocoder, with Laroche-Dolson phase-locking implemented on the same STFT substrate, which the near-perfect $\alpha = 1$ null on that grid confirms it tolerates. The per-segment dispatch — peak by default, heap on passages whose density masks its residual, seeded identically at each reset — applies each model only where the other is unacceptable, on a criterion of masking rather than dynamic. The cost of the approach is manual marker authoring and offline operation; in our evaluation, transparency on the demonstration material exceeds that of the general-purpose tools listed in Section 4.6.

References

[1] J. Laroche and M. Dolson, "Improved Phase Vocoder Time-Scale Modification of Audio," *IEEE Transactions on Speech and Audio Processing*, vol. 7, no. 3, pp. 323–332, May 1999. https://ieeexplore.ieee.org/document/759041

[2] Z. Prusa and N. Holighaus, "Phase Vocoder Done Right," in *Proc. 25th European Signal Processing Conference (EUSIPCO)*, 2017. https://ltfat.org/notes/ltfatnote050.pdf (Real-time phase gradient heap integration; building on Prusa & Sondergaard, DAFx-16, and Prusa, Balazs & Sondergaard, IEEE/ACM TASLP 25(5), 2017.)

Both methods are cited as integrated, not extended. The contribution of this work is the per-segment dual-model architecture built on a faithful PGHI substrate with Laroche-Dolson phase-locking on the shared STFT grid, and the per-section authoring findings on the target repertoire — not any modification to the underlying phase-correction methods themselves.